DIVIDE-AND-CONQUER AND DP PROBLEMS

1. Playing with Numbers Using DP

```
#include <stdio.h>

int main() {

int n;

scanf("%d", &n);

long long dp[n + 1];

dp[0] = 1;

for (int i = 1; i <= n; i++) {

dp[i] = dp[i - 1];

if (i >= 3) {

dp[i] += dp[i - 3];
```

```
}

}

printf("%lld", dp[n]);

return 0;

}
```

3. Longest Common Subsequence

```
#include <stdio.h>


#include<string.h>

int main() {

char s1[1000], s2[1000];

scanf("%s", s1);

scanf("%s", s2);

int n = strlen(s1);

int m = strlen(s2);

int dp[n + 1][m + 1];

for (int i = 0; i <= n; i++) {

for (int j = 0; j <= m; j++) {

if (i == 0 || j == 0) {

dp[i][j] = 0;
```

```
}

else if (s1[i - 1] == s2[j - 1]) {

dp[i][j] = dp[i - 1][j - 1] + 1;

}

else {

dp[i][j] = dp[i - 1][j] > dp[i][j - 1]

? dp[i - 1][j]

: dp[i][j - 1];

}

}

}

printf("%d", dp[n][m]);

return 0;

}
```

4. Longest Non-Decreasing Subsequence

```
#include <stdio.h>

int main() {

int n;

scanf("%d", &n);
```

```
int a[n];

int dp[n];

for (int i = 0; i < n; i++) {

scanf("%d", &a[i]);

dp[i] = 1;

}

int ans = 1;

for (int i = 1; i < n; i++) {

for (int j = 0; j < i; j++) {

if (a[j] <= a[i] && dp[j] + 1 > dp[i]) {

dp[i] = dp[j] + 1;

}

}

if (dp[i] > ans)

ans = dp[i];

}

printf("%d", ans);

return 0;

}
```